

Hüseyin Karaca

✉ huseyin.karaca@bilkent.edu.tr ☎ +90 530 307 3199 🏠 Bilkent 79, 06800, Ankara, Türkiye
🌐 huseyin-karaca.github.io in huseyinkaraca 🎓 Hüseyin Karaca 🔄 huseyin-karaca

Education

Master of Science in Electrical and Electronics Engineering <i>Bilkent University</i> Grade: 3.87/4.00 Advisor: Prof. Dr. Süleyman Serdar Kozat Thesis topic: Ensemble learning and feature selection methods for time-series forecasting	2024-07 – present Ankara, Türkiye
Bachelor of Science in Electrical and Electronics Engineering <i>Middle East Technical University</i> Grade: 3.60/4.00 Advisor: Assoc. Prof. Dr. Elif Vural Capstone Project: A smart monitor to keep track of vitals and sleep patterns of the baby.	2019-09 – 2024-07 Ankara, Türkiye
High School Diploma in Science <i>Ankara Science High School</i> Grade: 97.19/100	2015-09 – 2019-06 Ankara, Türkiye

Research Experience

Summer Intern <i>UCLA (The Ozcan Research Group)</i> Studied diffusion models for biomedical images under the supervision of Prof. Dr. Aydogan Ozcan. Helped the preparation of 3D printed diffractive layers for an all-optical image denoiser network.	2023-07 – 2023-09 Los Angeles, CA, USA
Undergraduate Researcher <i>METU (Dr. Elif Vural's Research Group)</i> Studied sample complexity of domain adaptation of different deep learning architectures. Performed regular neural network trainings with PyTorch to test novel theoretical bounds.	2022-09 – 2024-07 Ankara, Türkiye
Summer Intern <i>TUBITAK SAGE (Image Processing Division)</i> Studied on bad pixel detection and wrote a MATLAB code that simplifies the examination of calibration images. Improved robustness analysis algorithms by performing stitching on real seeker images.	2022-08 – 2022-09 Ankara, Türkiye
Undergraduate Researcher <i>METU (Communication Networks Group)</i> Studied age-aware random access for LoRa - LoRaWAN protocols. Employed a LoRaWAN network at the METU campus.	2021-07 – 2022-09 Ankara, Türkiye

Publications

- J3 **Karaca, H.**, Tumay, A., Koc, A.T., Kozat, S.S.. "Hierarchical ensemble-based feature selection for time series prediction". *Digital Signal Processing*, 2026. DOI: 10.1016/j.dsp.2025.105528
- J2 **Karaca, H.**, Kozat, S.S.. "Soft Gradient Boosting with Learnable Feature Transforms for Sequential Regression". *IEEE Signal Processing Letters*, 2024. DOI: 10.1109/LSP.2025.3639345
- J1 Isil, C., Gan, T., Ardic, F.O., Montesoglu, K., Digani, J., **Karaca, H.**, Chen, H., Li, J., Mengü, D., Jarrahi, M., Aksit, K., Ozcan, A.. "All-optical image denoising using a diffractive visual processor". *Light: Science & Applications*, 2024. DOI: 10.1038/s41377-024-01385-6
- C1 **Karaca, H.**, Akgul, O., Arslan, O.F., Aydemir, A.C., Aydin, F.S., Unsal, E.A., Vural, E.. "An Experimental Study of the Sample Complexity of Domain Adaptation". *IEEE Signal Processing and Communications Applications (SIU)*, 2023. DOI: 10.1109/SIU59756.2023.10223790
- P1 Vural, E., **Karaca, H.**. "A Unified Analysis of Generalization and Sample Complexity for Semi-Supervised Domain Adaptation". *arXiv preprint*, 2025. DOI: 10.48550/arXiv.2507.22632

Teaching Experience

Graduate Teaching Assistant

Bilkent University

Guided students during laboratory sessions, graded reports and quizzes, organized recitations.

- *EEE321 Signals and Systems (Fall 2024, Prof. Dr. Levent Onural).*
- *EEE321 Signals and Systems (Spring 2025, Prof. Dr. Billur Barshan).*
- *EEE321 Signals and Systems (Fall 2025, Spring 2026, Prof. Dr. Haldun Özaktas).*

2024-07 – present

Ankara, Türkiye

Undergraduate Teaching Assistant

Middle East Technical University

Assisted the preparation of lecture notes, organized monthly recitations.

- *EE201 Circuit Theory I (Fall 2023, Prof. Dr. Çağatay Candan).*
- *EE224 Electromagnetic Theory (Spring 2024, Prof. Dr. Gülbin Dural).*

2023-10 – 2024-06

Ankara, Türkiye

Skills

Languages:	Turkish (Native), English (C1), Persian (A2)
Programming:	PyTorch, Python, MATLAB, LaTeX, Scikit-learn, Nixtla, Sktime
Software:	Siemens NX, LTspice, Wordpress, Quartus II, Modelsim, Simulink
Hardware:	Oscilloscope, FPGA, SDR

Achievements and Honors

Research Awards

* ICT Authority of Türkiye 5G and Beyond Graduate Research Scholarship	10/25
* METU EEE STAR Departmental Undergraduate Research Program 3rd Place Award	06/22
* The Sci. Tech. Res. Council of Türkiye National Graduate Scholarship Program (2210-A)	09/24
* The Sci. Tech. Res. Council of Türkiye Scientific Activities Abroad Support Program (2224-D)	06/23

National Examinations

* Placed 24th among 2.6M participants in National University Entrance Examination (YKS)	07/19
* Placed 1st among 1.4M participants in National High School Entrance Examination (TEOG)	08/15

Physics Olympiads

* Bronze medalist in National Physics Olympiads	02/18
* Represented Turkey in Asian and European Physics Olympiads held in Russia and Estonia.	05/17
* The Sci. Tech. Res. Council of Türkiye Undergraduate Scholarship Program (2205)	09/19

Extracurricular Activities

Trekking and Camping

Participated in two trekking tours (200km) on the Ancient Lycian Way.

Organized midnight full moon walks and camps in Cappadocia.

Cycling

Board Member at METU Cycling Team

Represented METU in intercollegiate mountain and road bike races.

Organized weekly trainings and rides for the newcomers.

Student Council Presidency

Ankara Science High School

Organized a spring festival and "Happy New Year!" party (three years in a row).

Designed and sold printed hoodies for the Class of 2019.

Founded a book society and organized weekly book review meetings.